

Section 2.4

Partitioned Matrices

Partitioned Matrices

A partitioned matrix (or block matrix) is a matrix which is interpreted as having been divided into sections called blocks (or submatrices).

Example:

When we view a matrix as a collection of column vectors, we have partitioned the matrix into blocks (the vectors).

An Example of a Partitioned Matrix

The matrix

$$A = \begin{bmatrix} 1 & 4 & 2 & -1 & 3 \\ 5 & -2 & 3 & 1 & 1 \\ -6 & 4 & 1 & 2 & 9 \\ 7 & 1 & 3 & 1 & -5 \end{bmatrix}$$

can be partitioned as

$$A = \left[\begin{array}{cc|cc|c} 1 & 4 & 2 & -1 & 3 \\ 5 & -2 & 3 & 1 & 1 \\ -6 & 4 & 1 & 2 & 9 \\ 7 & 1 & 3 & 1 & -5 \end{array} \right]$$

Addition of Partitioned Matrices

If matrices A and B are the same size and are partitioned in exactly the same way, then the sum $A + B$ (which is also the same size) can also be partitioned the same way, and each block of $A + B$ can be computed separately as the sum of the corresponding blocks of A and B .

Multiplication of Partitioned Matrices

Partitioned matrices can be multiplied by the usual row-column rule as if the blocks were scalars, provided that the column partition of the first matrix matches the row partition of the second. Such partitions are said to be conformable for block multiplication.

Example 1

Assume the matrices are partitioned conformably for block multiplication. Compute the product.

$$\begin{bmatrix} I & 0 \\ -X & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Example 2

Find formulas for X , Y , and Z in terms of A , B , and C . Justify your calculations. In some cases, you may have to make assumptions about the size of a matrix to produce a formula.

$$\begin{bmatrix} A & B \\ C & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ X & Y \end{bmatrix} = \begin{bmatrix} 0 & I \\ Z & 0 \end{bmatrix}$$

The Column-Row Expansion of AB

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then,

$$AB = [\text{col}_1(A) \quad \text{col}_2(A) \quad \cdots \quad \text{col}_n(A)] \begin{bmatrix} \text{row}_1(B) \\ \text{row}_2(B) \\ \vdots \\ \text{row}_n(B) \end{bmatrix}$$
$$= \text{col}_1(A)\text{row}_1(B) + \cdots + \text{col}_n(A)\text{row}_n(B)$$

Each term in this expansion is called an outer product.

Example 3

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$.

Verify that AB can be calculated as $\text{col}_1(A)\text{row}_1(B) + \text{col}_2(A)\text{row}_2(B) + \text{col}_3(A)\text{row}_3(B)$

Inverses of Partitioned Matrices

A matrix of the form $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ is called block upper triangular.

The inverse of such a matrix is

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

Inverses of Partitioned Matrices

A block diagonal matrix is a partitioned matrix where all blocks which are not along the main diagonal of blocks are zero.

The inverse of a matrix

$$A = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}$$

is

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & A_{22}^{-1} \end{bmatrix}$$

Example 4

Without using row reduction, find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Numerical Notes

When matrices are too large to fit in a computer's high-speed memory, partitioning permits the computer to work with only a small number of submatrices (typically 2-3) at a time.

Professional software for high-performance numerical linear algebra makes intensive use of partitioned matrix calculations.

Numerical Notes

A research team was working on a problem which initially had 837 equations and over 12,750,000 variables. They were able to partition the resulting matrix into 837 rows and 51 block columns, each of which contained about 250,000 individual columns. This allowed them to solve the problem in about 4 minutes on a supercomputer.
